EI-HCR: An Efficient End-to-End Hybrid Consistency Regularization Algorithm for Semisupervised Remote Sensing Image Segmentation

Yizhe Xu, Liangliang Yan, and Jie Jiang

***Abstract*— Recently, remote sensing image (RSI) semantic segmentation technology has advanced greatly, with the fully supervised process achieving particularly strong performance. However, the technology depends heavily on dataset labels, leading to high annotation costs. To alleviate this problem, we propose a novel efficient end-to-end hybrid consistency regularization algorithm (EI-HCR) for the semisupervised semantic segmentation of RSI, wherein only a few labeled images and a large number of unlabeled images are effectively used. First, we devise data perturbation (DP) consistency regularization (CR), which includes a data mix-up method to combine unlabeled and labeled images. Then, we employ teacher and student networks to conduct model perturbation (MP) CR. Both segmentation results are regarded as pseudo-labels for each other. In the end, the semisupervised loss is composed of DP and MP consistency loss and supervises network training along with the fully supervised loss. More importantly, we first combine the characteristics of knowledge distillation to make the student network more lightweight, efficiently reducing the model inference time. Experimental results demonstrate the effectiveness of EI-HCR on the ISPRS Vaihingen and Massachusetts Buildings datasets. With only 5% of the labeled images, EI-HCR can achieve the same accuracy as the fully supervised training with 50% of the labeled images, and the number of student model parameters is only 9.64 M, indicating the method's great advantages over other algorithms.**

***Index Terms*— Consistency regularization (CR), knowledge distillation, remote sensing image (RSI), semantic segmentation, semisupervised learning.**

I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) is one of the key research areas in RSI processing. It involves pixel-level classification for various ground types and provides corresponding semantic information. It has been widely applied in land resource utilization [1], urban planning [2], and environment monitoring [3]. Traditional RSI methods are mostly based on thresholds [4], regional growth [5], watersheds [6], and other principles to extract low-level surface features. However, the final accuracy is unimpressive and semantic information is rarely mentioned. With the rapid development of deep learning, mainstream RSI methods are now mostly based on the convolutional neural network (CNN) and transformer [7], [8], [9] in which the encoder–decoder structures are designed to obtain various complex feature information and enhance the segmentation effect. Several typical network models are widely employed, such as UNet [10], Deeplab [11], [12], SegFormer [8], SETR [13], Swin Transformer [14], and so on. Although these methods above can achieve great accuracy, they also have limitations that must be addressed.

On the one hand, the deep learning technique is mostly data-driven and the model performance often relies heavily on dataset labels during training. The semantic segmentation of RSI is a dense prediction task and requires pixel-level labels, requiring professionals to spend much time and energy on dataset labeling. Compared with other visual tasks such as image classification and object detection, the annotation cost of RSI semantic segmentation is much higher. Moreover, unlabeled images often make up the majority of images in reality and are also easier to obtain. On the other hand, RSI has diverse object backgrounds, large-scale spans, and rich spectral characteristics, which bring greater challenges than ordinary scenes. Considering that the deep network has a stronger capability to extract feature information, many researchers usually adopt a complicated network to enhance the final segmentation effect. But this increases the number of parameters and the amount of calculation, bringing challenges to the actual application and deployment. Thus, it is critical to effectively utilize a large number of unlabeled images, reduce the dependence on the dataset labels, and make the model as lightweight as possible.

Many scholars have proposed a series of solutions to the problems caused by dataset labels, such as weak-supervised learning, domain adaptation, and semisupervised learning. The key point is how to effectively adopt a large amount of unlabeled data to further improve the model performance in the case of little labeled data. To reduce the reliance on the

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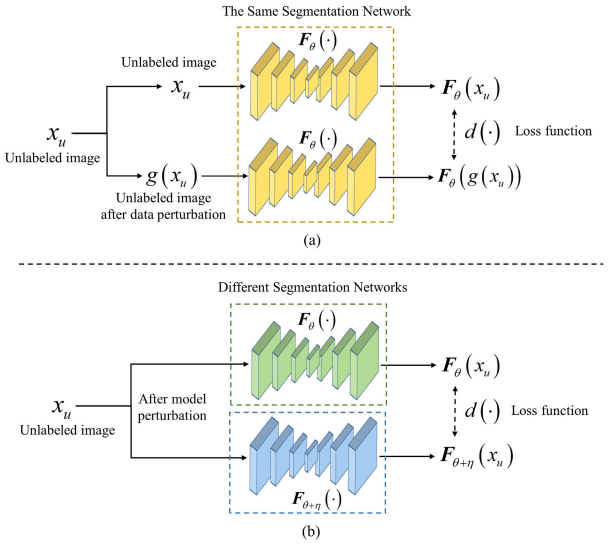


Fig. 1. Two perturbations of CR. (a) DP process. (b) MP process.

dataset annotation, weakly supervised semantic segmentation uses rough labels such as box-level or image-level labels and exploits some methods like class activation mapping (CAM) [15] or gradient-weighted CAM (Grad-CAM) [16] to obtain the corresponding class confidence maps. At the same time, it combines morphological operations, seed region growth, and other methods to conduct postprocessing to obtain the pixel-level pseudo-labels and then train the segmentation network. To a certain extent, the weak-supervised algorithm can decrease the dependence on dataset labels, but the pseudo-labels are too rough and the accuracy is unimpressive. In addition, many approaches utilize the generative adversarial network (GAN) [17], such as combining it with the CycleGAN [18] to make two different RSI datasets with similar data distribution or using discriminator results to generate the pseudo-labels for unlabeled images. However, GAN-based approaches are usually difficult to converge and have a high training cost. Semisupervised semantic segmentation can better balance the dataset label amount and model segmentation accuracy, and it is relatively convenient to implement. Most researchers design semisupervised algorithms based on consistency regularization (CR), including data perturbation (DP) and model perturbation (MP). Fig. 1 demonstrates the difference between these two perturbation modes. DP means that even if the same original image input is slightly disturbed, the final output result should be consistent with the original result. MP means that for the same original image, the corresponding prediction results of different segmentation networks should be as consistent as possible. Nowadays, CR-based methods can enhance the model performance with only a few labeled images, but the improvement effect is insignificant, and related studies have been inadequate. In addition, some researchers have mixed these two perturbation modes, but the training cost in their methods is high [19].

In this article, we propose an efficient semisupervised end-to-end hybrid CR algorithm, which is termed EI-HCR. The algorithm is based on the principle of CR and knowledge distillation. With only a few labeled images, EI-HCR effectively introduces a large number of unlabeled images and greatly improves the accuracy of the model. First, based on the improved data mix-up method, the algorithm combines the unlabeled image with the labeled image to generate the corresponding DP consistency loss. Then, a cross-pseudo-label supervision method comprising a teacher segmentation network and a student segmentation network is devised to generate the MP consistency loss. Furthermore, this pattern can help the two network branches learn from each other and enhance model generalization. Besides, the characteristics of knowledge distillation are utilized to make the student network lightweight, and further improve model accuracy. EI-HCR is an end-to-end algorithm that can directly output the segmentation results of unlabeled images, and calculate the corresponding fully-supervised and semisupervised loss to monitor the network training. Experimental results demonstrate the effectiveness of the proposed EI-HCR on the public International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Massachusetts Buildings datasets. With only 5% of the labeled samples, EI-HCR can obtain the same accuracy as the full-supervised training with 50% of the labeled samples, and the number of student model parameters is only 9.64 M, making the algorithm more efficient and less dependent on the dataset labels. The contributions of this article can be summarized as follows:

1) The data- and model-based perturbations are combined in our hybrid CR algorithm to enhance model generalization and achieve better performance for semisupervised segmentation in RSI.
2) Based on knowledge distillation, the actual segmentation network is extremely efficient and the complexity of the model is greatly reduced, which may speed up model inferencing.
3) The effectiveness of the proposed algorithm is demonstrated by extensive ablation and comparison experiments on different public datasets.

The rest of this article is organized as follows. Related work is introduced in Section II. EI-HCR and its components are detailed in Section III. The experimental settings and results are described in Section IV. The conclusion and prospects of the method are given in Section V.

II. Related Work

The semisupervised semantic segmentation of RSI has relatively lower requirements for dataset labels than the fully supervised approach, wherein each image is not forced to have a corresponding pixel-level label and no specific requirements for image-level annotation. Moreover, owing to the complex background and lack of labels, the semisupervised segmentation accuracy is usually lower, and related research has been inadequate compared with research on the fully supervised approach. Today, some advanced researchers have made some attempts to extract roads [20], buildings [21], and clouds [22] by performing semisupervised semantic segmentation methods. Originally, researchers mostly used simple pseudo-labels and self-training methods [23], [24]. However, the effectiveness of these methods depends heavily on the quality of pseudo-labels. To improve the quality of pseudo-labels, some reliable pseudo-labels selection methods have been gradually proposed [25]. In addition, some researchers design a unique

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contrast loss function to achieve semisupervised segmentation of RSIs [26]. Recently, in order to solve the problems of end-to-end training and generalization of models, GAN-based and CR-based methods have gradually become mainstream.

A. Based on Generative Adversarial Network

According to whether the domain-adaptive (DA) operation is carried out, the GAN-based methods of semisupervised semantic segmentation also include two parts. The DA GAN-based approaches focus on cross-domains of different datasets, wherein the source domain and target domain datasets are transformed into a similar data distribution and then the segmentation model performs well in different scenarios. Some common networks used for DA include CycleGAN [18], Cycada [27], and AdaptSegNet [28], and the main distinctions lie in domain adaptation in image space, feature space, and output space, respectively. Unlike DA methods, other GAN-based approaches conduct semisupervised research around a single data distribution. They employ the generator as the segmentation network and the discriminator to distinguish whether the input is an original label or the segmentation result. The discriminator outputs the confidence probability map and uses the trusted region by thresholds as the pseudo-label for the corresponding unlabeled image. Hung et al. [29] designed the semisupervised algorithm based on the GAN and adopted the fully connected network as the discriminator. Moreover, the adversarial loss and semisupervised loss are generated and supervise the training of the generator and discriminator with the labeled loss together. Rather than computing semisupervised loss, Souly et al. [30] treated the segmentation network as the discriminator and utilized the generator to expand the sample size of the dataset and promote the final segmentation effect.

Many attempts have been made to combine the GAN with the semisupervised segmentation of RSI. For example, Benjdira et al. [31] referred to the idea of CycleGAN and performed DA on RSIs. Compared with directly using source domain images, the accuracy after domain adaptation significantly improved. Liu et al. [32] designed a dual spatial domain adaptive network (BSANet), wherein the discrete wavelet transform is used for downsampling, and two discriminators are used to reduce the data distribution difference between the source domain and the target domain. Ji et al. [33] proposed a full-space domain adaptive network (FSDAN), which uses the synergistic image and feature adaptation (SIFA) [34] method to correct the data distribution difference between the source domain and target domain in image space and feature space respectively. Similar to Hung et al. [29] and Wang et al. [35] devised a semisupervised multiscale GAN for semisupervised segmentation of RSI, which integrates a GAN and the atrous spatial pyramid pooling (ASPP) [11] module. Sun et al. [36] further integrated the attention mechanism and the GAN and proposed the BAS^{4}Net semisupervised network, which consists of a boundary attention module and a channel-weighted module.

B. Based on Consistency Regularization

CR means that for the same input, data enhancement and network randomization will not change the final prediction results, including DP and MP. In semisupervised network training, for unlabeled images, it is advisable to adopt the CR to calculate the corresponding semisupervised loss. Yun et al. [37] proposed the CutMix semisupervised segmentation algorithm, combining two unlabeled images with masks by the cut-out [38] and mix-up [39] operations and then calculating the DP consistency loss to supervise model training. Inspired by CutMix, Olsson et al. [40] further studied and designed ClassMix, which integrates the mask layout with the segmented category distribution. Sohn et al. [41] devised FixMatch based on DP, wherein a weak data enhancement and strong data enhancement are added to unlabeled images, and the output of weak data enhancement is regarded as the pseudo-label to calculate the consistency loss with the output of strong data enhancement. Tarvainen and Valpola [42] proposed the classic mean teacher (MT) algorithm based on MP, wherein only the teacher network executes gradient backpropagation and the student network parameters are updated from the previous iteration process using the exponential moving average (EMA) algorithm. Ke et al. [43] proposed the guided collaborative training (GCT) semisupervised method by inputting the same image into two networks and calculating the mean squared error (MSE) loss or Kullback-Leibler (KL) divergence between both confidence probability maps as the consistency loss. Chen et al. [44] employed cross pseudo-label supervision (CPS) in semisupervised training. Similar to MT [42], CPS also uses teacher and student networks, but these two networks in CPS have the same structure and their segmentation results are treated as each other's pseudo-labels. Recently, Liu et al. [19] designed a semisupervised algorithm (PSMT) that combines perturbed and strict MTs. They employed multiple networks to supervise each other and devised weighted cross-entropy loss and various perturbations to improve the model accuracy. However, multiple networks are also used in the inference process, which may significantly increase the inference time.

Research has also been done on applying the CR to semisupervised RSI segmentation. For example, Xia et al. [45] combined the DP to generate pseudo-labels, designed the SDLED network, and utilized it in semisupervised building footprint generation. Based on FixMatch [41], Wang et al. [46] devised the CRAUP and adopted the strategy of automatically updating pseudo-labels and expanding the capacity of the dataset to train the network. Nevertheless, the entire training process consists of multiple procedures to continuously enhance the quality of pseudo-labels, and it is not an end-to-end approach. Furthermore, they also designed the RanPaste [47] network which combines the characteristics of MT and CutMix and uses thresholds to enhance the confidence regions of the network output. He et al. [48] discussed the applications of CR in semisupervised RSI segmentation in detail, and they proposed the ClassHyPer semisupervised algorithm. However, they only adopted a simple fully convolutional network (FCN) [49] network as the baseline for experiments without discussing the effectiveness of the algorithm for different networks. Wang et al. [50] upgraded FixMatch and devised ICNet so that two networks are trained, and the identities of the teacher network and student network are constantly changed in the training process, which may cause current

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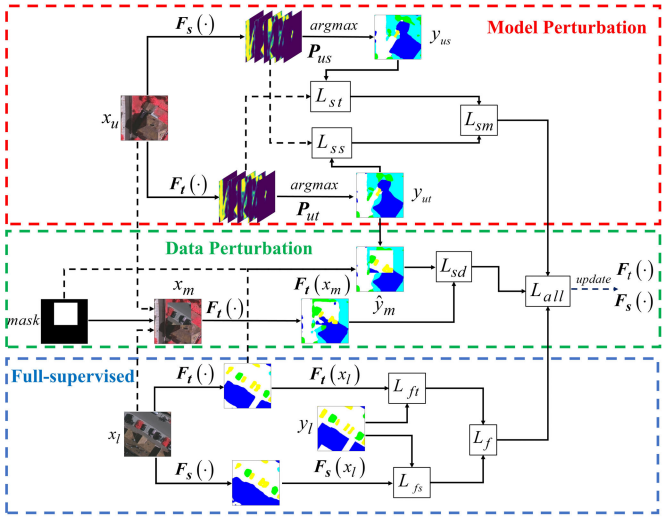


Fig. 2. Overall structure of EI-HCR semisupervised algorithm, including MP, DP, and full-supervised process.

network parameters to not be updated promptly during the backpropagation process. Moreover, Zhang et al. [51] adopted a variety of DP CR methods and exploited the approach of MT to train networks. However, they did not distinguish the teacher and student network or consider the model complexity.

Overall, both GAN-based and CR-based semisupervised methods can alleviate the network dependence for pixel-level labels. However, GAN-based methods have the disadvantage of unstable training, which easily leads to the nonconvergence of the model. While existing CR-based methods often require large training costs, such as initializing multiple large-scale teacher networks [19] or adopting complex DP methods [48], requiring multiple calculations and preservation of the results of the forward propagation and reverse gradient propagation. Thus, the existing semisupervised training process is still cumbersome and the segmentation accuracy also has room for further improvement. To improve the generalization of the model and fully explore the advantages of CR, we combined two perturbation modes. Our EI-HCR carried out model-level perturbation based on teacher-student network architecture and realized double-branch cross-learning. Meanwhile, by performing an improved mix-up method in the teacher network branch, unlabeled data and labeled data are fused and stable pseudo-labels are generated. Thus, effective data level perturbation is also achieved. Finally, considering that the existing semisupervised methods often have high training costs, the knowledge distillation is applied to lightweight student networks, which is beneficial to reduce the training overhead of our semisupervised models.

III. Methodology

EI-HCR for semisupervised segmentation of RSI is shown in Fig. 2 and consists of three parts: MP, DP, and a fully supervised process. To facilitate the following elaboration, let us specify some symbols accordingly. The labeled dataset is marked $D_l = (x_l^k, y_l^k)$: $1 \le k \le N$; the corresponding number of labeled images is N, (x_l^k, y_l^k) denotes the kth original image and its label, respectively. The unlabeled dataset is marked $D_u = (x_u^k)$: $1 \le k \le M$, the number of images is M, and x_u^k denotes the kth unlabeled image. In the actual scene, M is far greater than N. The idea of EI-HCR is to combine labeled dataset and unlabeled dataset to conduct semisupervised training on the segmentation network, which can achieve better results than fully supervised training that only uses a labeled dataset. In addition, there are two networks in the process of MP and distillation learning. Here, the teacher segmentation network is masked $\boldsymbol{F_t}(\cdot)$, and the student segmentation network is masked $\boldsymbol{F_s}(\cdot)$. The loss function in the training process is denoted as L and different losses are expressed with different subscripts. For example, the total loss is L_{all}, the semisupervised loss is L_s, and the teacher network semisupervised loss is L_{st}.

A. DP Consistency Regularization

In the process of DP, EI-HCR adopts the data mix-up operation to effectively combine the unlabeled image x_u and labeled image x_l. First, we use a mask to cut part of x_l and paste it to the same position of x_u, and then we obtain the mixed image x_m

$$x_m = \text{mask} \odot x_l + (1 - \text{mask}) \odot x_u \tag{1}$$

where the mask is a binary image (each value of a pixel is 0 or 1) and $\odot$ denotes the dot-product operation. Additionally, the mask area shape that value equals one is square-shaped, and the length is 0.5 times the input image size by default.

Then, both x_l and x_u are input into the pretrained teacher network $\boldsymbol{F_t}(\cdot)$, and we get the corresponding segmentation results $\boldsymbol{F_t}(x_l)$ and $\boldsymbol{F_t}(x_u)$. Similarly, we also employ the same mask to cut part of $\boldsymbol{F_t}(x_l)$ and paste it in the same position of $\boldsymbol{F_t}(x_u)$ and the outcome y_m is regarded as a pseudo-label for mixed image x_m

$$y_m = \text{mask} \odot \boldsymbol{F}_t(x_l) + (1 - \text{mask}) \odot \boldsymbol{F_t}(x_u). \tag{2}$$

Finally, the x_m is input into $\boldsymbol{F_t}(\cdot)$ to get the result $\boldsymbol{F_t}(x_m)$ and the DP semisupervised loss L_{sd} is calculated with the pseudo-label y_m

$$L_{\text{sd}} = H(F_t(x_m), y_m) \tag{3}$$

where $H(\cdot)$ denotes the loss function.

It can be seen from the above process that the DP of EI-HCR combines labeled and unlabeled images with a mask and produces a mixed image to calculate the semisupervised loss. If two unlabeled images are directly used to perform the data mix-up operation, the segmentation model will not learn the spatial feature distribution of the labeled dataset D_l and unlabeled dataset D_u. This is because the network gradients are effectively updated on the D_l, and the pseudo-labels for the mixed image from D_u are ambiguous. If the objects are replaced in the data mix-up operation, which means using a mask to cut part of x_u and paste it in the same position of x_l, we find that the final effect decreases greatly. One explanation is that the segmentation network will be more inclined to learn the feature distribution of D_l with a few labeled images. The D_u containing a large number of unlabeled images cannot be effectively used, so the network can easily be overfitting.

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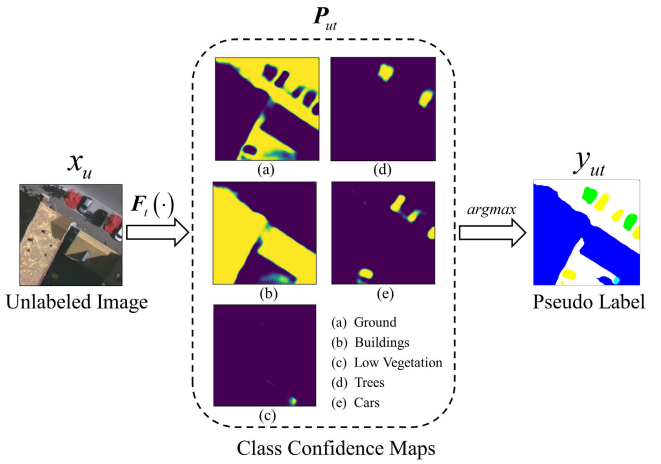


Fig. 3. Process of generating pseudo-label based on class confidence maps. (a)–(e) Class confidence map results of ground, buildings, low vegetation, trees, and cars, respectively.

B. Model Perturbation CR

In the MP process, EI-HCR utilizes two different and independent segmentation networks to conduct cross-pseudo-supervision, which is closely linked with the DP in the previous step to form a hybrid CR. As shown in Fig. 2, we input the unlabeled image x_u into the teacher network $\boldsymbol{F_t}(\cdot)$ and student network $\boldsymbol{F_s}(\cdot)$, and then we attain the corresponding confidence probability maps $\boldsymbol{P}_{\text{ut}}$ and $\boldsymbol{P}_{\text{us}}$. Afterward, we take the argmax operation and obtain the pseudo-labels y_{ut} and y_{us}

$$\begin{cases} x_u \rightarrow \boldsymbol{P}_{\text{ut}} \rightarrow y_{\text{ut}} \\ x_u \rightarrow \boldsymbol{P}_{\text{us}} \rightarrow y_{\text{us}}. \end{cases} \tag{4}$$

Fig. 3 depicts the process of generating pseudo-label y_{ut} when unlabeled image x_u is input into the teacher network $\boldsymbol{F_t}(\cdot)$. Furthermore, the pseudo-label y_{ut} generated from $\boldsymbol{F_t}(\cdot)$ and the segmentation result $\boldsymbol{F_s}(x_u)$ are used to calculate the consistency loss L_{ss}, while the pseudo-label y_{us} generated from $\boldsymbol{F_s}(\cdot)$ and segmentation result $\boldsymbol{F_t}(x_u)$ are used to calculate the consistency loss L_{st}. The loss function is multiclassification cross-entropy. The total MP semisupervision loss L_{sm} is the sum of L_{ss} and L_{st}

$$L_{\text{ss}} = -\frac{1}{h \cdot w} \sum_{i=1}^{h} \sum_{j=1}^{w} \sum_{k=1}^{c} y_{\text{ut}}^{ijk} \log\left(F_s(x_u)^{ijk}\right) \tag{5}$$

$$L_{\text{st}} = -\frac{1}{h \cdot w} \sum_{i=1}^{h} \sum_{j=1}^{w} \sum_{k=1}^{c} y_{\text{us}}^{ijk} \log\left(F_t(x_u)^{ijk}\right) \tag{6}$$

$$L_{\text{sm}} = L_{\text{ss}} + L_{\text{st}} \tag{7}$$

where h and w denote the length and width of the input image, respectively; c denotes the number of categories.

Some scholars have employed other loss functions, such as L1 regularization and MSE, to calculate the MP CR loss. However, they are equivalent to making the output probability values of both network outputs as consistent as possible. In fact, the distributions of the output confidence probability values of different models are different. Comparatively, by applying the argmax operation, generating the pseudo-label, and using the cross-entropy function to calculate the loss, we can ensure that both models effectively learn the same distribution of the original dataset, and this approach can better meet the requirements of CR. If a single segmentation network is directly used and perturbations are imposed on it, such as adding multiple decoders or using the EMA algorithm to update another network parameters, even though cross pseudo-supervision can be adopted later, only one network is constantly learning and iterating in essence. Thus, the capability of feature extraction is lower than the two mutually independent teacher and student segmentation networks.

C. Efficient End-to-End Training

Generally, a large model is often a single complex network or a collection of several networks, which has higher performance and accuracy, but the number of parameters is often large, and the inference speed is slow. A small model always has limited expression ability but fewer parameters and a faster inference speed. Knowledge distillation aims to use the knowledge learned by the large model to guide the training of a small model so that the latter has the same performance as the former model. In addition, the large model is often called the teacher network, and the small model is called the student network. The proposed EI-HCR combines the characteristic of knowledge distillation to make the student segmentation network as lightweight as possible. The method of efficient end-to-end training in EI-HCR is shown in Fig. 4.

End-to-end means that just by inputting the unlabeled image x_u, labeled image x_l and mask, the teacher segmentation network $\boldsymbol{F_t}(\cdot)$ and the student segmentation network $\boldsymbol{F_s}(\cdot)$ can output the corresponding semantic segmentation results, calculate the full supervision loss and semisupervision loss, and back propagate and update the network parameters. There is no need to specifically stop the network training to generate pseudo-labels and other procedures. The knowledge distillation operation embodies in the process of MP, wherein the student network uses the lightweight model with m convolution layers, and the teacher network uses a more complex model with n convolution layers. By default, m is far less than n. Finally, only the student network is used for inference and prediction after training. Moreover, a few labeled images are also input into both $\boldsymbol{F_t}(\cdot)$ and $\boldsymbol{F_s}(\cdot)$, and we calculate the fully supervised loss L_f with the corresponding labels y_l. The semisupervised loss L_s is the sum of the DP loss L_{sd} and MP loss L_{sm}, and it is weighted to form the total loss L_{all} with the fully supervised loss

$$L_f = H(F_t(x_l),\ y_l) + H(F_s(x_l),\ y_l) \tag{8}$$

$$L_s = L_{\text{sd}} + L_{\text{sm}} \tag{9}$$

$$L_{\text{all}} = L_f + \lambda L_s \tag{10}$$

where $H(\cdot)$ denotes the loss function and in this article, we use the cross-entropy function. λ is the weighted factor of semisupervised loss and in later experiments we will discuss it in detail.

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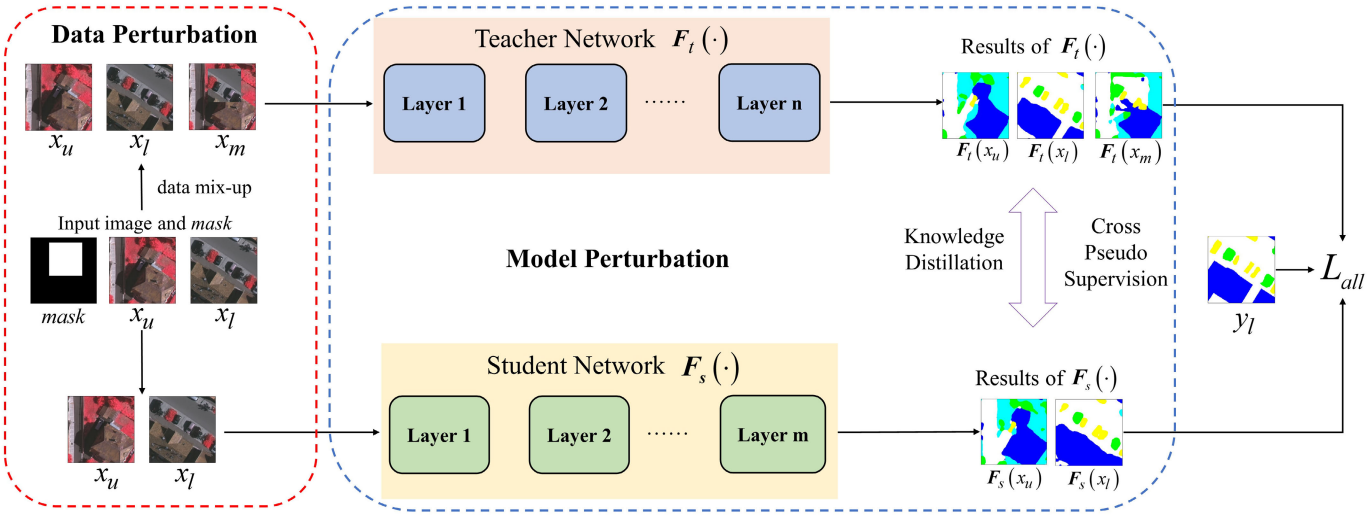


Fig. 4. Efficient and end-to-end training, in which the knowledge distillation is adopted in MP.

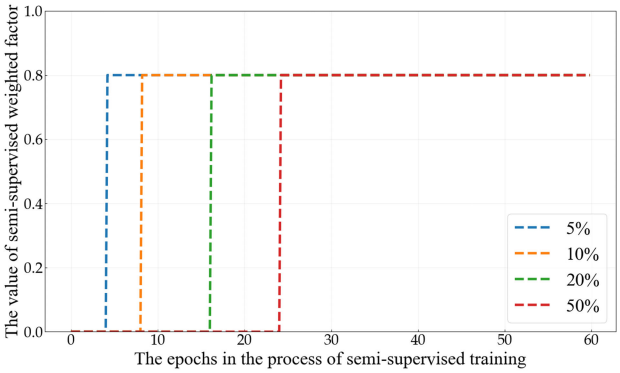


Fig. 5. Pretrained warm-up process with different proportions of labeled images.

D. Pretrained Warm-Up

In order to enable the model to have certain prior knowledge when introducing a large number of unlabeled images and avoid falling into overfitting with a small number of labeled images, the EI-HCR algorithm devises a simple but useful pretrained warm-up method, as shown in Fig. 5. The whole process is similar to a step function. During the training process, the segmentation models are pretrained separately after some epochs on the labeled dataset. At this time, we still calculate the semisupervised loss but set the weighted factor λ to 0. After reaching the number of pretrained epochs, the weighted factor is not 0 and the semisupervision loss is introduced to supervise the network training. Moreover, different proportions of the labeled dataset in the whole dataset correspond to different numbers of pretrained warm-up epochs. We set the proportions (5%, 10%, 20%, and 50%) corresponding to pretrained epochs (4, 8, 16, and 24) by default.

IV. Experiments

In this section, a series of experiments are designed to verify the corresponding effect of the semisupervised algorithm EI-HCR. First, we introduce the experimental details, including the used datasets, data preprocessing methods, model training settings, and precision evaluation metrics. Then, we present the ablation experimental results to prove the effectiveness of subunits in the EI-HCR, which includes the selections of the semisupervised weighted factor, pretrained warm-up training, different perturbations of CR, different segmentation model baselines, and effect of knowledge distillation. Furthermore, we demonstrate the comparison of experimental results with other semisupervised segmentation algorithms on different datasets. Finally, we discuss the model complexity to testify to the lightweight of our algorithm.

A. Implementation Details

1) Datasets: In this article, experiments are conducted on two public datasets, including the ISPRS Vaihingen dataset and the Massachusetts Buildings dataset. The Vaihingen dataset, released in the 2-D semantic label challenge held by ISPRS, consists of six common ground categories. The names and colors of each category are: ground (white); buildings (blue); low vegetation (light blue); trees (green); cars (yellow); and clutter (red, also known as background). There are 33 images in the dataset, with the image size ranging from 1996 $\times$ 1995 to 3816 $\times$ 2550. The dataset is divided as in the ISPRS semantic challenge where 16 images are used for training and validation, and 17 images are only used for testing. The clutter category is not considered in the final accuracy calculation and boundary-eroded ground-truth labels are used. The Massachusetts Buildings dataset, composed of aerial images of Boston, was released in 2013. It mainly includes various buildings in urban and suburban areas, such as personal houses and garages. Each image size is 1500 $\times$ 1500, and the number of ground object categories is two, corresponding to buildings and backgrounds respectively. It is often used for building segmentation and extraction of urban RSIs; 141 images are selected for training and validation and ten images are selected for testing. Both datasets adopt the method of combining offline data enhancement with online data enhancement. Online data enhancement mainly

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includes the random rotation of specified angles (0°, 90°, 180°, and 270°), random vertical and horizontal flipping, and data normalization. The data normalization includes the mean factor, which is set to (0.4119, 0,4251, and 0.4327) and the variance factor which is set to (0.2741, 0.2851, and 0.2828). Offline data enhancement refers to cropping a large image into small patches for suitable model input and the patch size was 256×256.

2) Training Settings: Considering the large-scale variation of RSI, we employ pretrained HRNet_w18 [52] as the student network and pretrained HRNet_w32 [52] as the teacher network because feature maps with different resolutions are parallel connected and constant multiscale feature fusion is operated in the above networks. All semisupervised algorithm is built with the Pytorch deep learning framework. Stochastic gradient descent with momentum (SGDM) is employed as the optimizer and the GPU platform is NVIDIA GTX 3090 (24 GB RAM). In addition, the learning rate is initialized as 0.005 and adjusted with the poly learning rate attenuation strategy. The momentum is set to 0.9, and the batch size is set to eight. The number of training epochs is 200.

3) Evaluation Metrics: The evaluation metrics include overall accuracy (OA), intersection over union (IoU), mean IoU (mIoU), precision (P), recall (R), and F1-score. For the convenience of understanding, we assume that there are k ground categories, and P_{ij} denotes the number of pixels that belong to category i but are predicted as category j. The corresponding calculation formulas are as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{k} P_{ii}}{\sum_{i=1}^{k}\sum_{j=1}^{k} P_{ij}} \quad \mathrm{IoU}_i = \frac{P_{ii}}{\sum_{j=1}^{k} P_{ij} + \sum_{j=1}^{k} P_{ji} - P_{ii}} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{k}\sum_{i=1}^{k} \frac{P_{ii}}{\sum_{j=1}^{k} P_{ij} + \sum_{j=1}^{k} P_{ji} - P_{ii}} \quad P_i = \frac{P_{ii}}{\sum_{j=1}^{k} P_{ji}} \tag{12}$$

$$R_i = \frac{P_{ii}}{\sum_{j=1}^{k} P_{ij}} \quad F1_i = \frac{2P_i R_i}{P_i + R_i} \quad \mathrm{mF1} = \frac{1}{k}\sum_{i=1}^{k} F1_i. \tag{13}$$

B. Ablation Study

EI-HCR combines the principle of CR and knowledge distillation and effectively utilizes a large number of unlabeled images to train the segmentation network. Moreover, the network is supervised by weighting the semisupervised loss with supervised loss. Therefore, the value of the weighted factor and the effect of different perturbations need to be comprehensively explored in the ablation experiments. At the same time, in order to introduce extensive unlabeled images, EI-HCR adopts the method of the pretrained warm-up, which is also supposed to be proven feasible. In addition to the current segmentation network backbone HRNet, several other common backbones, such as Deeplabv3+, are selected as the student network and teacher network baselines to verify the robustness of EI-HCR. All ablation experiments are conducted on the ISPRS Vaihingen and Massachusetts Buildings datasets and the adopted proportions of labeled images are (5%, 10%, 20%, and 50%), respectively. The precision evaluation indexes include mIoU, the average F1 score mF1, and the overall precision OA.

TABLE I
ABLATION EXPERIMENTAL RESULTS (%) FOR THE SEMISUPERVISED WEIGHTED FACTOR WITH 5% AND 20% OF LABELED IMAGES ON THE VAIHINGEN DATASET

λ	5% (221)			20% (885)		
	mIoU	mF1	OA	mIoU	mF1	OA
0	77.02	86.79	88.32	80.68	89.16	89.96
0.4	81.03	89.37	90.31	82.41	90.22	90.81
0.6	80.57	89.05	90.04	82.50	90.28	90.80
0.8	**81.57**	**89.70**	90.53	**82.77**	**90.44**	**90.95**
1.0	81.49	89.65	**90.59**	82.76	**90.44**	90.92

TABLE II
ABLATION EXPERIMENTAL RESULTS (%) FOR THE SEMISUPERVISED WEIGHTED FACTOR WITH 5% AND 20% OF LABELED IMAGES ON THE MASSACHUSETTS BUILDINGS DATASET

λ	5% (189)			20% (756)		
	mIoU	mF1	OA	mIoU	mF1	OA
0	75.07	84.99	90.88	78.76	87.56	92.58
0.4	78.87	87.63	92.68	80.80	88.93	93.43
0.6	79.20	87.89	92.58	81.03	89.08	**93.59**
0.8	**79.71**	**88.21**	**93.00**	**81.09**	**89.12**	93.55
1.0	79.60	88.13	92.94	81.07	**89.12**	93.51

1) Effect of Semisupervised Weighted Factor Value: Tables I and II illustrate the quantitative experimental results corresponding to the change in the semisupervised weighted factor λ with 5% and 20% of labeled images on the Vaihingen and Massachusetts Buildings datasets. Here we set the value of the weighted factor in (0, 0.4, 0.6, 0.8, and 1.0), and λ equals 0 means that we only employ the provided labeled images to conduct fully supervised training. The bold values in the tables indicate the best results. Clearly, compared with the results of full-supervised training, the segmentation accuracy is significantly higher when λ is not equal to 0 and 0.8 is the best value. With 5% of the labeled images, λ equals to 0.8 results in a 4.55% improvement in mIoU, a 2.91% improvement in mF1, and a 2.21% improvement in OA on the Vaihingen dataset. On the Massachusetts Buildings dataset, mIoU, mF1, and OA are also improved by 4.64%, 3.22%, and 2.12%, respectively.

As shown in Fig. 6, as the value of the weighted factor gradually increases, all the evaluation indexes also gradually increase until they reach the maximum, wherein λ is around 0.8. Also, once the value has started to increase, the accuracy metrics begin to decline. This changing trend is consistent with the actual situation. At first, with the increase of the weighted factor, a large number of unlabeled images are introduced. During the process of network training, it is no longer limited to extracting feature information from a small number of labeled images, but it is feasible to integrate the extensive characteristics of unlabeled images, thus reducing the risk of model overfitting. However, with the further increase of the semisupervised weighted factor, the proportion of unlabeled image loss in the total loss increases. Actually, the labels used to calculate the semisupervised loss are not

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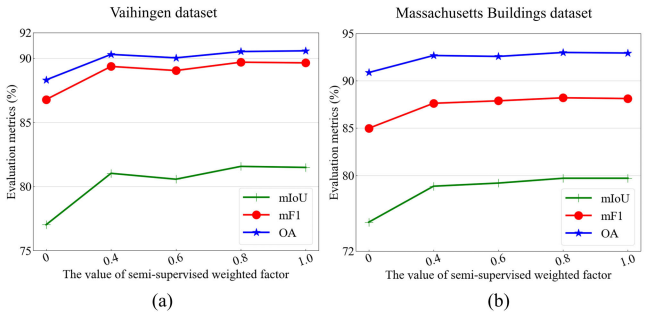


Fig. 6. Change curves between the semisupervised weighted factor and evaluation metrics with 5% of labeled images on different datasets. (a) Vaihingen dataset. (b) Massachusetts Building dataset.

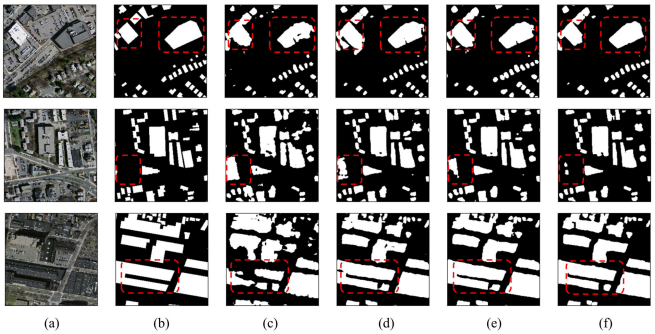


Fig. 7. Qualitative experimental results of different values of the semisupervised weighted factor λ on the Massachusetts Buildings dataset. (a) Input image. (b) Ground truth. (c) $\lambda = 0$. (d) $\lambda = 0.4$. (e) $\lambda = 0.6$. (f) $\lambda = 0.8$.

real labels but pseudo-label generated from hybrid CR. The quality of pseudo-labels is strongly affected by the model performance. If the semisupervised loss weight is too large, it will influence the effective constraints of the fully supervised loss for the model, thereby limiting the further improvement of the model accuracy. Therefore, we set the final weight factor to 0.8. Fig. 7 demonstrates qualitative results given the change in the weighted factor on the Massachusetts Buildings dataset. With the increase of semisupervised loss weight, the segmentation effect of dense buildings is significantly improved, and errors of small blocks in large-scale buildings decline. Additionally, some confusing ground objects, such as playgrounds in a color similar to buildings, can also be effectively distinguished.

2) Effect of Different Perturbations: Tables III and IV describe the ablation results of adopting the different perturbations of EI-HCR with 5% and 20% of labeled images on the Vaihingen and Massachusetts Building datasets. It can be seen that adding DP or MP alone can improve the model segmentation accuracy to a certain extent. Especially, our algorithm EI-HCR can effectively integrate the features of the above two perturbations and further enhance the performance.

With 20% of labeled images, EI-HCR combining DP and MP achieves 2.09%, 1.28%, and 0.99% improvement in mIoU, mF1, and OA, respectively on the Vaihingen dataset, and 2.33%, 1.56%, and 0.97%, respectively on the Massachusetts Building dataset. Fig. 8 depicts the feature map information of different categories for the effect of different perturbations. When there are features such as irregular terrain and shadows

TABLE III

ABLATION EXPERIMENTAL RESULTS (%) OF DIFFERENT PERTURBATIONS WITH 5% OF LABELED IMAGES ON THE VAIHINGEN AND MASSACHUSETTS BUILDINGS DATASETS

DP	MP	Vaihingen dataset			Massachusetts Buildings dataset		
		mIoU	mF1	OA	mIoU	mF1	OA
×	×	77.02	86.79	88.32	75.07	84.99	90.88
✓	×	78.13	87.52	88.75	76.07	85.67	91.59
×	✓	80.64	89.12	90.19	78.69	87.48	92.75
✓	✓	**81.57**	**89.70**	**90.53**	**79.71**	**88.21**	**93.00**

TABLE IV

ABLATION EXPERIMENTAL RESULTS (%) OF DIFFERENT PERTURBATIONS WITH 20% OF LABELED IMAGES ON THE VAIHINGEN AND MASSACHUSETTS BUILDINGS DATASETS

DP	MP	Vaihingen dataset			Massachusetts Buildings dataset		
		mIoU	mF1	OA	mIoU	mF1	OA
×	×	80.68	89.16	89.96	78.76	87.56	92.58
✓	×	81.55	89.70	90.18	79.62	88.15	92.86
×	✓	81.97	89.95	90.67	80.72	88.86	93.53
✓	✓	**82.77**	**90.44**	**90.95**	**81.09**	**89.12**	**93.55**

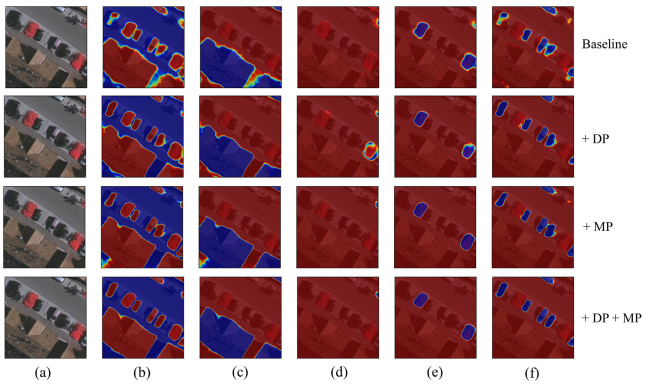


Fig. 8. Visualization of different category feature maps under different perturbations on the Vaihingen dataset. (a) Input image. (b) Ground. (c) Buildings. (d) Low vegetation. (e) Trees. (f) Cars.

inside the RSI, the category feature drift is more obvious near the building and car from the baseline results. After the addition of either the MP or DP CR introduced in this article, the above effect gradually declines. What's more, after the addition of the EI-HCR semisupervised training method, more useful information from a large number of unlabeled images is extracted and the feature drift effect is reduced particularly.

3) Effect of Pretrained Warm-Up: Table V describes the ablation results considering whether adopting the pretrained warm-up with different proportions of labeled images on the Vaihingen dataset. It is evident that with 5% of labeled images, all the precision metrics are improved after using the pretrained warm-up. The mIoU, mF1, and OA are increased by 1.16%, 0.75%, and 0.55%, respectively. Table VI lists the results with different proportions of labeled images on the Massachusetts Buildings dataset. Even if the mIoU and mF1 are reduced slightly, with 10% of labeled images, the segmentation accuracy is further improved with 5%, 20%, and 50% of labeled images. In addition, as the proportion

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TABLE V
ABLATION EXPERIMENTAL RESULTS (%) FOR PRETRAINED WARM-UP WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE VAIHINGEN DATASET

Pretrained warm up	Proportions of labeled images	mIoU	mF1	OA
×	5%	80.41	88.95	89.98
×	10%	82.01	90.01	90.58
×	20%	82.61	90.32	90.93
×	50%	83.01	90.55	91.01
✓	5%	**81.57**	**89.70**	**90.53**
✓	10%	**82.29**	**90.15**	**90.71**
✓	20%	**82.77**	**90.44**	**90.95**
✓	50%	**83.05**	**90.61**	**91.03**

TABLE VI
ABLATION EXPERIMENTAL RESULTS (%) FOR PRETRAINED WARM-UP WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE MASSACHUSETTS BUILDINGS DATASET

Pretrained warm up	Proportions of labeled images	mIoU	mF1	OA
×	5%	78.74	87.52	92.78
×	10%	**80.61**	**88.82**	93.26
×	20%	81.06	**89.12**	93.43
×	50%	81.92	89.68	93.80
✓	5%	**79.71**	**88.21**	**93.00**
✓	10%	80.58	88.77	**93.42**
✓	20%	**81.09**	**89.12**	**93.55**
✓	50%	**82.04**	**89.75**	**93.86**

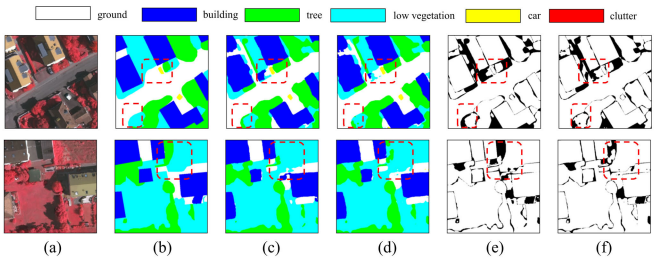


Fig. 9. Qualitative experimental results considering whether pretrained warm-up is adopted with 5% of labeled images on the Vaihingen dataset. (a) Input image. (b) Ground truth. (c) Segmentation results of no pretrained warm-up. (d) Segmentation results of pretrained warm-up. (e) Binary error images of no pretrained warm-up. (f) Binary error images of pretrained warm-up.

of labeled images increases, the improvement effect brought by pretrained warm-up gradually weakens, which is actually in line with expectation. Because after the number of labeled samples increases, the network can obtain more feature information from the existing labeled dataset. Even if the pseudo-labels generated by CR for unlabeled images are of low quality at this time, the model can still converge effectively under the supervision of labeled data. Fig. 9 visualizes the qualitative comparison results on the Vaihingen dataset with 5% of labeled images. It can be observed that for some confusing categories in RSI, such as grass and trees, or the ground and buildings, it is easy to make mistakes without the pretrained warm-up. In fact, the method of the pretrained warm-up can help the segmentation network learn prior knowledge in advance and prepare well for using a large number of unlabeled images later. As a result, the segmentation accuracy for easily confusing ground objects can be improved.

4) Effect of Different Student and Teacher Network Baselines: In addition to the HRNet [52] used as the student network baseline, we also adopt several other classical segmentation networks for semisupervised experiments, such as UNet [10], Deeplabv3+ [12], and PSPNet [53]. These networks employ ResNet50, which has been pretrained on the ImageNet dataset as the backbone. Further, apart from the student network baseline, all the other experimental settings remain unchanged. Table VII demonstrates the quantitative results of fully supervised experiments without EI-HCR and semisupervised experiments with EI-HCR on the Vaihingen dataset. Compared with the fully supervised results, it is apparent that the accuracy of different student networks increases significantly after using EI-HCR. With 5% of labeled images, the evaluation index mIoU of UNet increases from 75.87% to 80.17%, an improvement of 4.3%. The mIoU of Deeplav3+ increases from 75.27% to 80.44%, an improvement of 5.17%. And the mIoU of PSPNet increases from 75.37% to 80.15%, an improvement of 4.78%. Even if the number of labeled images rises, with 20% of labeled images, each baseline still shows a significant improvement with EI-HCR. The mIoU of UNet has increased from 79.47% to 82.03%, an improvement of 2.56%. The mIoU of Deeplav3+ increases from 79.98% to 81.48%, an improvement of 1.5%. And the mIoU of PSPNet increases from 79.15% to 81.66%, an improvement of 2.51%.

In addition, Table VIII lists the quantitative results of the Massachusetts Buildings dataset, and there are also clear improvements. With 5% of labeled images, the mIoU of UNet, Deeplabv3+, and PSPNet increases by 1.53%, 3.48%, and 0.52%, respectively. Also, with 20% of labeled images, the mIoU of UNet, Deeplabv3+, and PSPNet increases by 1.59%, 3.07%, and 1.73%, respectively. In the above discussion, except for mIoU, other evaluation indexes, like mF1 and OA, are also improved. Besides, we also select Deeplabv3+ with the pretrained ResNet-101 as the teacher network and replace HRNet_w18 with different student networks. Table IX describes the related results on the Vaihingen dataset with 5% and 20% of labeled images. Obviously, even if another teacher network is adopted, compared to directly using a small number of labels for fully supervised experiments, there is a significant promotion in various evaluation indexes after adding the EI-HCR semisupervised algorithm. With 5% of labeled images, the mIoU of student networks UNet, Deeplabv3+, PSPNet, and HRNet are improved by 3.74%, 4.19%, 3.61%, and 4.03%, respectively, indicating that our semisupervised algorithm can alleviate the label dependency problem and achieve high accuracy with as few labels as possible.

Fig. 10 describes the qualitative ablation experimental results on the Massachusetts Buildings dataset. With only a few numbers of labeled images for fully supervised training, UNet, Deeplabv3+, and PSPNet have a poor segmentation effect on buildings that are large-scale and densely distributed. Particularly, when the color of buildings is close to the ground, the building blocks are directly stuck together in the final segmentation results. After the EI-HCR algorithm has been added

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TABLE VII

ABLATION EXPERIMENTAL RESULTS (%) CONSIDERING WHETHER EI-HCR IS ADOPTED FOR DIFFERENT STUDENT NETWORK BASELINES (HRNET AS TEACHER NETWORK) WITH 5% AND 20% OF LABELED IMAGES ON THE VAIHINGEN DATASET

EI-HCR	Backbone	5% (189)			20% (756)		
		mIoU	mF1	OA	mIoU	mF1	OA
×	UNet	75.87	86.06	87.80	79.47	88.39	89.49
×	Deeplabv3+	75.27	85.60	87.70	79.98	88.33	89.54
×	PSPNet	75.37	85.67	87.80	79.15	88.18	89.47
×	HRNet	77.02	86.79	88.32	80.68	89.16	89.96
✓	UNet	**80.17**	**88.83**	**90.00**	**82.03**	**89.94**	**90.77**
✓	Deeplabv3+	**80.44**	**89.00**	**90.13**	**81.48**	**89.64**	**90.54**
✓	PSPNet	**80.15**	**88.80**	**90.14**	**81.66**	**89.76**	**90.59**
✓	HRNet	**81.57**	**89.70**	**90.53**	**82.77**	**90.44**	**90.95**

TABLE VIII

ABLATION EXPERIMENTAL RESULTS (%) CONSIDERING WHETHER EI-HCR IS ADOPTED FOR DIFFERENT STUDENT NETWORK BASELINES (HRNET AS TEACHER NETWORK) WITH 5% AND 20% OF LABELED IMAGES ON THE MASSACHUSETTS BUILDINGS DATASET

EI-HCR	Backbone	5% (221)			20% (885)		
		mIoU	mF1	OA	mIoU	mF1	OA
×	UNet	74.82	84.74	91.20	78.50	87.39	92.47
×	Deeplabv3+	72.36	82.90	90.22	76.32	85.83	91.80
×	PSPNet	69.45	80.59	89.09	74.99	84.89	91.14
×	HRNet	75.07	84.99	90.88	78.76	87.56	92.58
✓	UNet	**76.35**	**85.82**	**92.01**	**80.09**	**88.44**	**93.32**
✓	Deeplabv3+	**75.84**	**85.46**	**91.73**	**79.39**	**87.96**	**93.06**
✓	PSPNet	**69.97**	**80.92**	**89.75**	**76.72**	**86.11**	**91.96**
✓	HRNet	**79.71**	**88.21**	**93.00**	**81.09**	**89.12**	**93.55**

TABLE IX

ABLATION EXPERIMENTAL RESULTS (%) CONSIDERING WHETHER EI-HCR IS ADOPTED FOR DIFFERENT STUDENT NETWORK BASELINES (DEEPLABV3+ AS TEACHER NETWORK) WITH 5% AND 20% OF LABELED IMAGES ON THE VAIHINGEN DATASET

EI-HCR	Backbone	5% (189)			20% (756)		
		mIoU	mF1	OA	mIoU	mF1	OA
×	UNet	75.87	86.06	87.80	79.47	88.39	89.49
×	Deeplabv3+	75.27	85.60	87.70	79.98	88.33	89.54
×	PSPNet	75.37	85.67	87.80	79.15	88.18	89.47
×	HRNet	77.02	86.79	88.32	80.68	89.16	89.96
✓	UNet	**79.61**	**88.47**	**89.79**	**81.64**	**89.75**	**90.55**
✓	Deeplabv3+	**79.46**	**88.35**	**89.73**	**81.93**	**89.93**	**90.80**
✓	PSPNet	**78.98**	**88.01**	**89.94**	**81.70**	**89.79**	**90.62**
✓	HRNet	**81.05**	**89.37**	**90.57**	**82.51**	**90.28**	**90.81**

for semisupervised training, the segmentation errors mentioned above are significantly decreased. In fact, the extracted features information of a small number of the labeled images cannot serve as an overview of the whole feature distribution of the dataset, and the segmentation networks easily cause overfitting. When we combine the segmentation network with the EI-HCR algorithm, a large number of unlabeled images are introduced effectively so that the network can extract more abundant feature information, which then improves the final segmentation effect.

5) Effect of Knowledge Distillation: To confirm the effectiveness of the knowledge distillation method in EI-HCR, we also provide experimental results of the teacher network (HRNet_w32) on the Vaihingen and Massachusetts Buildings datasets. As shown in Tables X and XI, the accuracy of the student network is close to that of the teacher network. Especially, with 5% of labeled images, the accuracy index of the student network is the same or higher. Among them, the OA index of the student network on the Vaihingen dataset is 0.1% higher than that of the teacher network, and mIoU and mF1 on the Massachusetts Buildings dataset have also increased by 0.31% and 0.24%, respectively. However, as the labeled sample size gradually increases, the accuracy of teacher networks is gradually higher than that of student networks. In fact, this situation is in line with expectations, because in EI-HCR, the teacher network and student network combine each other's pseudo-labels to conduct knowledge distillation. When the number of labeled samples is very small, the teacher network is more complex and easier to get overfitting. At this time, the effect of the student network is close to or even more than that of the teacher network. As the proportion of labeled images increases, complex teacher

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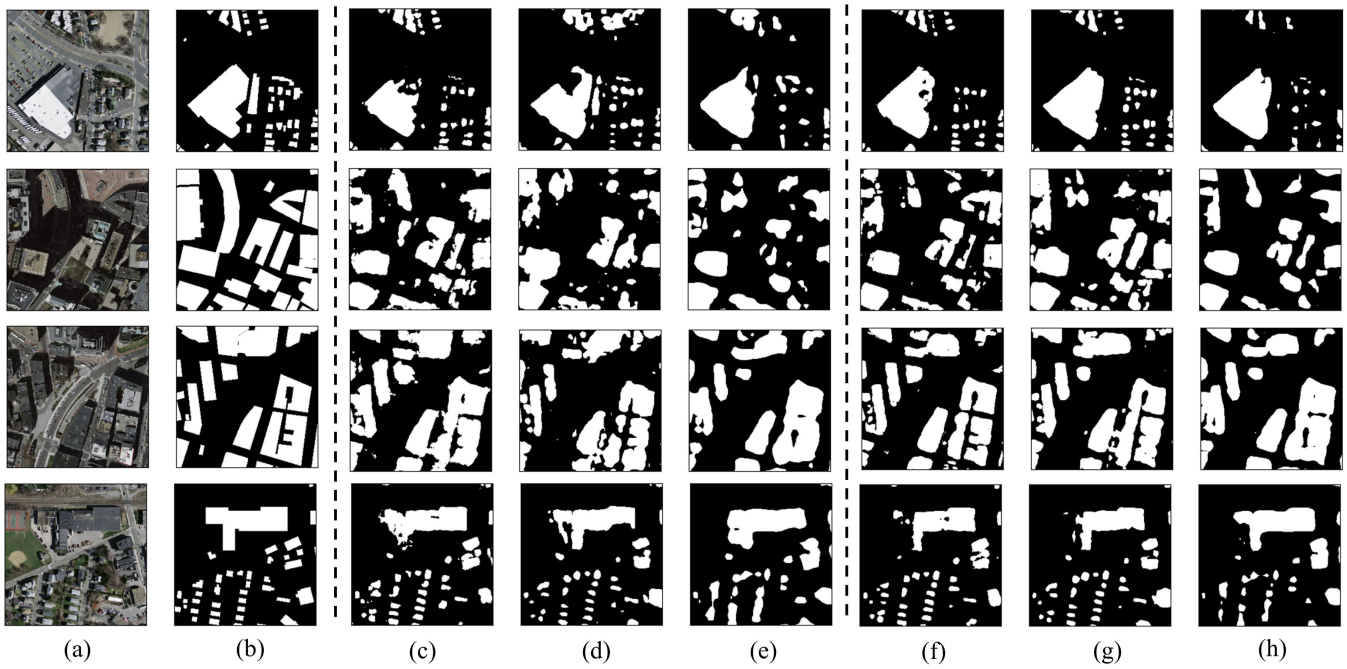


Fig. 10. Visualization results considering whether using the EI-HCR semisupervised algorithm for different student network baselines with 5% and 20% of labeled images on the Massachusetts Buildings dataset. (a) Input image. (b) Ground truth. (c) UNet. (d) Deeplabv3+. (e) PSPNet. (f) EI-HCR + UNet. (g) EI-HCR + Deeplabv3+. (h) EI-HCR + PSPNet.

TABLE X
EXPERIMENTAL RESULTS (%) FOR TEACHER AND STUDENT NETWORKS WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE VAIHINGEN DATASET

Proportions	Teacher Network HRNet_w32			Student Network HRNet_w18		
	mIoU	mF1	OA	mIoU	mF1	OA
5%	**81.57**	89.69	90.41	**81.57**	**89.70**	**90.53**
10%	**82.59**	**90.34**	**90.90**	82.29	90.15	90.71
20%	**83.02**	**90.59**	**91.16**	82.77	90.44	90.95
50%	**83.19**	**90.70**	**91.19**	83.05	90.61	91.03

TABLE XI
EXPERIMENTAL RESULTS (%) FOR TEACHER AND STUDENT NETWORKS WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE MASSACHUSETTS BUILDINGS DATASET

Proportions	Teacher Network HRNet_w32			Student Network HRNet_w18		
	mIoU	mF1	OA	mIoU	mF1	OA
5%	79.40	87.97	**93.03**	**79.71**	**88.21**	93.00
10%	**81.10**	**89.13**	**93.56**	80.58	88.77	93.42
20%	**81.63**	**89.48**	**93.71**	81.09	89.12	93.55
50%	**82.51**	**90.04**	**94.19**	82.04	89.75	93.86

networks can learn more features, gradually improving accuracy while efficiently distilling the student network, allowing the student network to further improve and be as close as possible to the teacher network.

C. Comparisons With Other Methods

1) Results on Vaihingen Dataset: The quantitative results of EI-HCR and other existing CR-based semisupervised algorithms on the Vaihingen dataset are presented in Table X. The baseline method means that only a few labeled images are used for fully supervision training and that a large number of unlabeled images are not used. Among other semisupervised algorithms, FixMatch [28] and CutMix [26] are based on DP; MT [29], GCT [30], and CPS [31] are based on MP; and PSMT [19] is based on mixed perturbations. The results vary with the proportion of labeled and unlabeled images. To be fair and consistent, all of the methods adopt the same settings as EI-HCR, such as the segmentation network baseline, model super-parameters, and dataset batch size.

As shown in Table XII, EI-HCR achieves the highest accuracy when the proportions of labeled images are 5%, 10%, 20%, and 50%; particularly, when the number of labeled images is small, the performance of EI-HCR is significantly improved compared with that of other methods. With 5% of labeled images, the accuracy shows a 1.24% increase in mIoU, a 0.79% increase in OA, and a 0.75% increase in mF1 compared with the second-best PSMT semisupervised algorithm. Furthermore, each evaluation index is close to the fully supervised baseline results with 50% of labeled images. Fig. 11 shows the qualitative experimental results of different semisupervised algorithms. It can be observed that in some scenes where buildings and trees, as well as grass and trees, are intertwined, the MT and FixMatch methods are prone to uneven fracturing of building segmentation results and confusion between grass and trees. EI-HCR employs hybrid CR to calculate the consistency loss of DP and MP for a large number of unlabeled images so that the segmentation network can be supervised more effectively during training and the segmentation error can be reduced. Additionally, the student network interacts with and supervises the teacher network to further extract effective information from complex networks during the training process.

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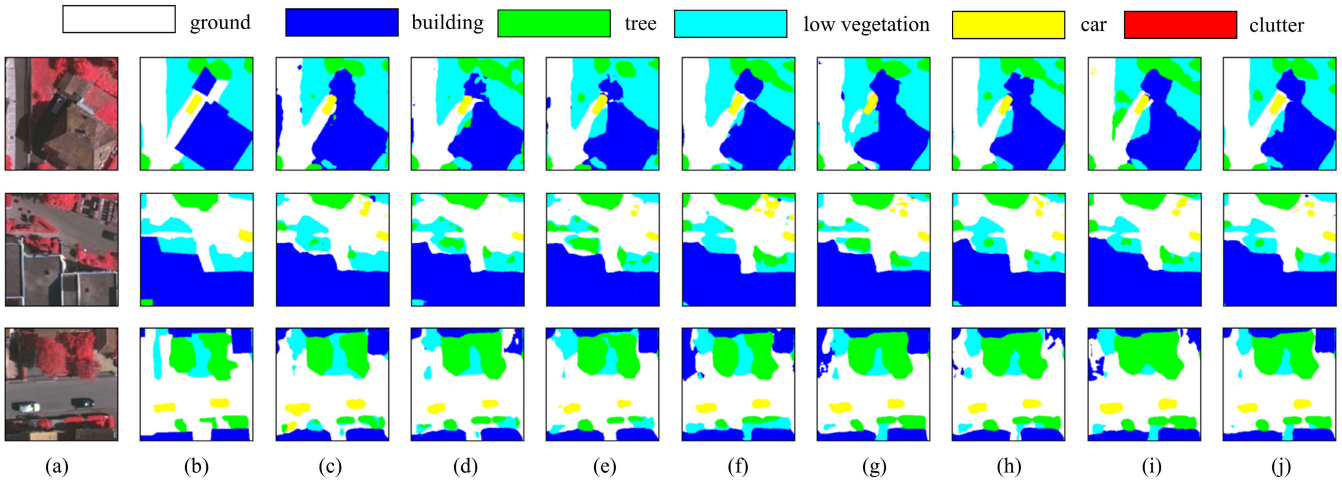


Fig. 11. Visualization results of different semisupervised algorithms on the Vaihingen dataset. (a) Input image. (b) Ground truth. (c) Baseline. (d) MT. (e) GCT. (f) FixMatch. (g) CutMix. (h) CPS. (i) PSMT. (j) EI-HCR.

TABLE XII
COMPARISON OF EXPERIMENTAL RESULTS OF DIFFERENT SEMISUPERVISED ALGORITHMS WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE VAIHINGEN DATASET

Vaihingen dataset	5% (221)			10% (442)			20% (885)			50% (2213)		
	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA
Baseline[52]	77.02	86.79	88.32	79.37	88.33	89.29	80.68	89.16	89.96	81.99	89.96	90.49
MT[42]	77.37	87.01	88.44	79.52	88.41	89.41	81.40	89.60	90.22	81.91	89.92	90.55
GCT[43]	77.42	87.06	88.41	79.73	88.56	89.58	81.05	89.39	89.98	82.27	90.14	90.64
FixMatch[41]	78.98	88.06	89.14	80.51	89.06	89.73	81.63	89.74	90.33	81.76	89.82	90.54
CutMix[37]	79.58	88.46	89.22	81.33	89.57	90.07	82.26	90.14	90.56	82.41	90.22	90.61
CPS[44]	79.88	88.63	89.57	81.62	89.74	90.49	82.42	90.23	90.77	83.03	90.6	**91.04**
PSMT[19]	80.33	88.91	89.78	81.44	89.61	90.24	82.21	90.10	90.66	82.53	90.29	90.85
EI-HCR	**81.57**	**89.70**	**90.53**	**82.29**	**90.15**	**90.71**	**82.77**	**90.44**	**90.95**	**83.05**	**90.61**	91.03

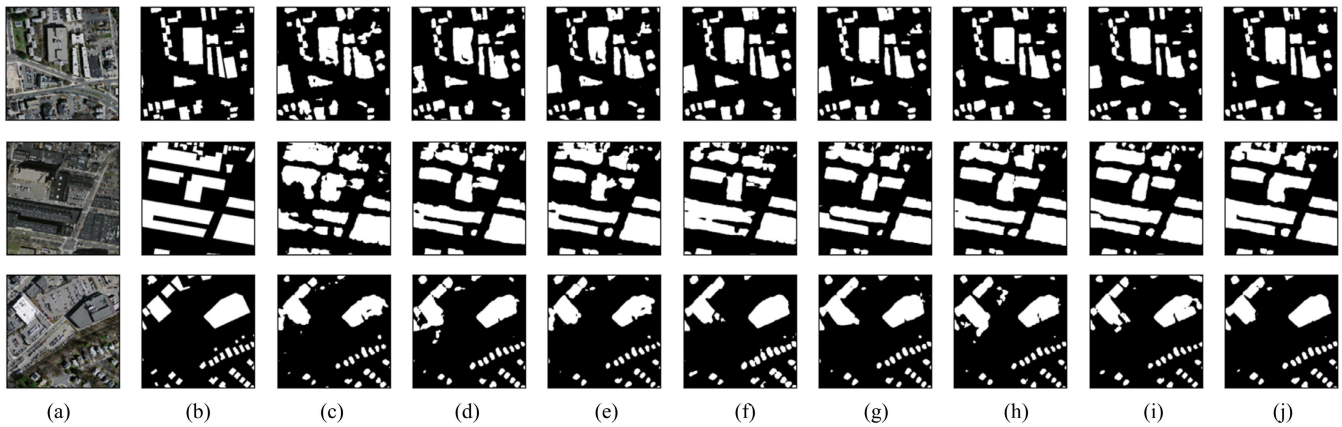


Fig. 12. Visualization results of different semisupervised algorithms on the Massachusetts Buildings dataset. (a) Input image. (b) Ground truth. (c) Baseline. (d) MT. (e) GCT. (f) FixMatch. (g) CutMix. (h) CPS. (i) PSMT. (j) EI-HCR.

2) Results on Massachusetts Buildings Dataset: Table XIII demonstrates the quantitative results with different proportions of labeled images on the Massachusetts Buildings dataset. The evaluation accuracy metrics of EI-HCR are also better than those of other methods under the proportions of 5%, 10%, and 50%. EI-HCR also has the highest OA with 20% of labeled images. With a small number of labeled images, EI-HCR also performs extremely well. When the proportion of labeled samples is 5%, EI-HCR improves by 0.65% in mIoU and 0.47% in mF1 compared with the second-best PSMT algorithm. Fig. 12 intuitively shows the qualitative experimental results of different methods. For large-scale buildings and

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TABLE XIII
COMPARISON EXPERIMENTAL RESULTS OF DIFFERENT SEMISUPERVISED ALGORITHMS WITH DIFFERENT PROPORTIONS OF LABELED IMAGES ON THE MASSACHUSETTS BUILDINGS DATASET

Massachusetts	5% (189)			10% (378)			20% (756)			50% (1890)		
Buildings dataset	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA
Baseline[52]	75.07	84.99	90.88	77.17	86.44	92.06	78.76	87.56	92.58	80.75	88.90	93.37
GCT[42]	76.07	85.67	91.56	78.28	87.23	92.42	79.30	87.92	92.83	81.26	89.24	93.59
MT[43]	76.40	85.89	91.78	78.20	87.15	92.56	79.53	88.07	92.98	80.69	88.85	93.47
FixMatch[41]	77.84	86.94	92.08	79.10	87.80	92.70	79.87	88.32	93.02	80.99	89.08	93.31
CutMix[37]	78.53	87.40	92.50	79.78	88.26	92.95	80.45	88.71	93.24	81.98	89.72	93.81
CPS[44]	78.63	87.46	92.62	80.30	88.61	93.17	**81.17**	**89.20**	93.36	81.89	89.66	93.76
PSMT[19]	79.06	87.74	92.90	79.95	88.34	93.26	80.89	88.98	93.50	81.72	89.53	93.80
EI-HCR	**79.71**	**88.21**	**93.00**	**80.58**	**88.77**	**93.42**	81.09	89.12	**93.55**	**82.04**	**89.75**	**93.86**

TABLE XIV
COMPARISON OF MODEL COMPLEXITY FOR DIFFERENT SEGMENTATION BASELINES

Model	FLOPs(G)	Params(M)	FPS
UNet	54.93	90.06	1.07
PSPNet	42.67	43.32	19.33
Deeplabv3+	44.37	46.58	15.87
Teacher Network	11.34	29.55	12.80
Student Network	**4.63**	**9.64**	**21.34**

other ground objects, the segmentation results of EI-HCR are smoother and there are less errors of fewer stripes than those of other algorithms. Besides, EI-HCR also achieves better effects for complex scenes such as those with building shadows in the image.

Furthermore, it is obvious that as the proportion of labeled images increases, the improvement brought by the above semisupervised algorithms gradually weakens. The promotion is particularly significant when the number of labeled images is very small, which indicates that our approach is also an effective way to avoid overfitting. In fact, it is worth mentioning that the distribution of a large amount of unlabeled data should be as consistent as possible with the provided small amount of labeled data. Otherwise, when there is a significant difference in the distribution of unlabeled and labeled images, the quality of generated pseudo-labels may not meet the requirements, which would lead to difficulty in convergence or low accuracy of the model.

D. Model Complexity

In addition to the metrics used to evaluate the segmentation accuracy, model complexity is also used to demonstrate the efficiency of our algorithm. As the student network in EI-HCR is employed for prediction and inference, it is necessary to compare the model complexity of the EI-HCR student network with that of other segmentation models. Similar to previous ablation experiments, we also adopt UNet, PSPNet, and Deeplabv3+ for comparison, wherein the pretrained ResNet50 is also chosen as the backbone. Moreover, the teacher network of EI-HCR is also included and the results are displayed in Table XIV. The model complexity consists of spatial complexity and computational complexity, and we utilize the number of model parameters (Params), floating point operations per second (FLOPs), and frames per second (FPS) as the evaluation indices. Additionally, the calculation of FLOPs is affected by the size of the input image, which we set to 256×256.

As shown in Table XIV, the model complexity of the EI-HCR student network shows a significant decrease compared with other segmentation models. The FLOPs of the EI-HCR student network are 4.63 G, which is the lowest result and denotes a faster inference speed. Also, the number of parameters of the EI-HCR student network is 9.64 M, which is about 20.7% of the parameters of Deeplabv3+ and 32.6% of those of the EI-HCR teacher network. Furthermore, the FPS of the EI-HCR student network amount to 21.34, which is the highest result and means that predicting one 256×256 px image takes only 47.86 ms. The above results align with the actual situation because we apply the knowledge distillation in the MP CR in semisupervised training of EI-HCR so that the student network can be as lightweight as possible.

V. CONCLUSION

Today, fully supervised methods for RSI segmentation can achieve high accuracy, but most depend heavily on dataset labels, leading to high annotation costs in practical applications. To solve the above problems, we have proposed EI-HCR in this article. This algorithm combines the principle of hybrid CR and knowledge distillation, effectively introduces a large number of unlabeled images with a small number of labeled images, greatly improves the segmentation accuracy, and makes the segmentation network more lightweight. The whole semisupervised training process is end-to-end, meaning that the segmentation result can be obtained by directly inputting the image to be segmented. The semisupervised loss is calculated for the unlabeled image and monitors the network training together with the full-supervised loss after being weighted by the corresponding factor. Experiments have been carried out on the multicategory dataset ISPRS Vaihingen and the binary dataset Massachusetts Buildings. In the ablation experiments, the value of the semisupervised weight factor, the effects of different perturbations, pretrained warm-up, different student and teacher network baselines, and knowledge

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distillation have been examined in detail. Especially, with only 5% of the labeled images, EI-HCR has achieved the same accuracy as the fully supervised training with half the labeled images. Moreover, extensive comparison experiments with other semisupervised methods have been conducted to verify the effectiveness of the EI-HCR algorithm.

In the future, three areas can be further studied. First, reinforcing the structure of the teacher and student networks with the transformer's multihead attention mechanism may obtain more global features of RSI. Second, delving into the ad hoc interpretability of CR algorithms to elucidate precisely how the segmentation model works. Finally, combining multisource RSI such as hyperspectral and SAR images may improve the generalization ability of the segmentation network.

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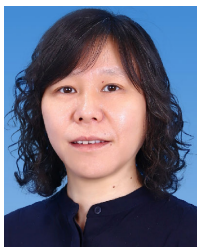
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